

Doppler-Free Two-Photon Modulation Transfer Spectroscopy in Sodium Dimers[★]

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Abstract. We performed an experiment to investigate modulation transfer spectroscopy in two-photon transition of Na₂ using a CW dye laser. Comparing with UV fluorescence detection, the signals obtained by modulation transfer spectroscopy have a very high S/N without the Doppler background problem and its line shape is in agreement with theoretical calculations.

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Optical heterodyne spectroscopy is very sensitive to weak absorption and is applied to saturation absorption and two-photon Doppler-free spectroscopy [1–5]. Using amplitude-modulated heterodyne polarization spectroscopy, the intrinsic sensitivity of Doppler-free polarization spectroscopy can be enhanced toward the short-noise limit [6]. One of the particularly interesting cases is modulation transfer spectroscopy where the optical heterodyne detection is performed via nearly-degenerate four-wave mixing (NDFWM) in the nonlinear medium [7–11]. This technique has very high sensitivity without the Doppler background problem. It has been applied to stabilizing lasers and precision spectroscopy measurement [12–14]. During the past years, it has been extended from saturation absorption to two-photon transition in atoms [15] and the three-level quantum system in ions [16]. Recently, we became very interested in the uses of modulation transfer spectroscopy to study two-photon transitions of molecules with optical phase modulation. A pump beam is modulated by an optical phase modulator at the frequency δ . The phase modulation produces

sindbands which are located symmetrically around the laser optical frequency ω . The modulated pump beam collimates the counter-propagating unmodulated probe beam in the nonlinear medium. Via NDFWM at resonance of the nonlinear medium, the re-emitted electromagnetic fields propagate in the direction of the original, unmodulated probe beam. As a results, the new sindbands are generated and located around the probe beam at the frequencies $\omega \pm \delta$. It obvious that the intensity of new sindbands is independent of the Doppler background. The generation of new sindbands demonstrates that the modulation can be transferred from the modulated pump beam to the original, unmodulated probe beam via NDFWM in the nonlinear medium. When a fast photodiode is used, optical heterodyne beating occurs between the weak new sindbands and the probe beam which has enough power to be a local optical oscillator. The frequency of beating is same as the optical phase modulation frequency δ . Thus the spectrum's information is encoded at the phase modulation frequency which is sufficiently high to avoid excess laser amplitude noise that is dominant at low frequencies [3, 14]. This technique makes detection sensitivity close to the short-noise limit. In fluorescence detection, an efficient collector of fluorescence photons has to be constructed, and scattered light has to be carefully reduced. In modula-

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tion transfer spectroscopy, the signal is proportional to the length of the interaction region, which is sufficiently long to detect the two-photon transition with weak fluorescence. The scattered light from optical phase modulated pump beam makes no contribution to the signal because sidebands on the pump beam are located symmetrically around the laser optical frequency that produces beating at δ to cancel on the surface of photodiode [2, 3]. In this paper, we describe an experiment to investigate modulation transfer spectroscopy in two-photon transitions of Na_2 using a CW dye laser. In the tuning range of a DCM dye, we observed several two-photon transitions from ground single states to high-lying triple states in Na_2 . Comparing with UV fluorescence detection, the signals have

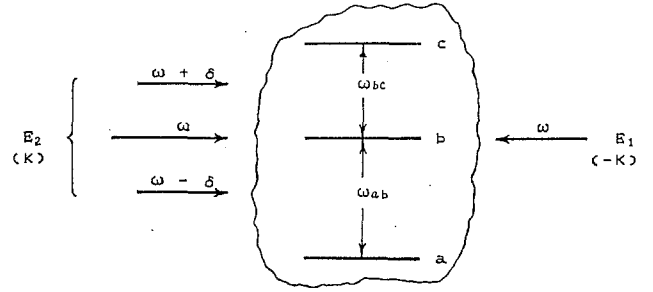


Fig. 1. Cascade three-level model for calculating the NDFWM response with optical phase modulation

high S/N without Doppler background due to the near resonance of intermediate energy level, and its line shape is in agreement with theoretical calculation.

1. Line Shape of Two-Photon Modulation Transfer Spectroscopy

For two-photon transition, cascade three-level quantum system (Fig. 1) is considered as the nonlinear medium. The density matrix equations of motion describing this system is given by

$$\left(\frac{\partial}{\partial t} + v \frac{\partial}{\partial z}\right) \rho_{ij} = r_i n_i - r_{ij} \rho_{ij} - i(\omega_i - \omega_j) \rho_{ij} + \frac{i}{\hbar} \sum_k E(\mu_{ik} \rho_{kj} - \rho_{ik} \mu_{kj}), \quad (1)$$

where $i, j = a, b$, and c .

The field of the pump beam modulated by an optical phase modulator at frequency δ is expressed by

$$E_2 = E_{20} [J_0 e^{i\omega t} + J_1 e^{i(\omega+\delta)t} - J_1 e^{i(\omega-\delta)t}] e^{-ikz} + \text{c.c.}, \quad (2)$$

where J_0 and J_1 are Bessel functions.

The field of the probe beam is given by

$$E_1 = E_{10} e^{i\omega t} \cdot e^{ikz} + \text{c.c.} \quad (3)$$

Eight possible combinations listed in the Table 1 can regenerate the field of E_r in the direction of $(-k)$ with the frequency $\omega \pm \delta$. Using perturbation theory and the rotating-wave approximation, the third-order elements of the density matrix described in (1) are given by [17]

$$\begin{aligned} \rho_{ab}^{(3)}(\omega \pm \delta) = & \mp (i/\hbar)^3 N \frac{E_{10} E_{20}^2 J_0 J_1 \mu_{ab}}{r_{ab} + i(\Delta_{ab} \pm \delta + kv)} \\ & \times \left[\sum_j \frac{\mu_{ab}^2}{r_j \pm i\delta} \left(\frac{1}{r_{ab} + i(\Delta_{ab} \pm \delta - kv)} + \frac{1}{r_{ab} - i(\Delta_{ab} - kv)} - \frac{1}{r_{ab} + i(\Delta_{ab} - kv)} - \frac{1}{r_{ab} - i(\Delta_{ab} \mp \delta - kv)} \right) \right. \\ & + \frac{\mu_{bc}^2}{r_{ac} + i(\Delta_{ac} \pm \delta)} \left(\frac{1}{r_{ab} + i(\Delta_{ab} \pm \delta - kv)} + \frac{1}{r_{ab} + i(\Delta_{ab} + kv)} \right) - \frac{\mu_{bc}^2}{r_{ac} + i\Delta_{ac}} \left(\frac{1}{r_{ab} + i(\Delta_{ab} - kv)} \right. \\ & \left. \left. + \frac{1}{r_{ab} + i(\Delta_{ab} + kv)} \right) \right] \times e^{i[(\omega \pm \delta)t + kz]}, \quad (4) \end{aligned}$$

Table 1. Eight possible combinations for regenerating the field of E_r in the direction of $(-k)$ with the frequency $\omega \pm \delta$

ω_l	k_l	ω_m	k_m	ω_n	k_n	ω_r	k_r
$\omega \pm \delta$	k	ω	k	ω	$-k$	$\omega \pm \delta$	$-k$
ω	$-k$	ω	k	$\omega \pm \delta$	k	$\omega \pm \delta$	$-k$
ω	k	$\omega \mp \delta$	k	ω	$-k$	$\omega \pm \delta$	$-k$
ω	$-k$	$\omega \mp \delta$	k	ω	k	$\omega \pm \delta$	$-k$

$$\omega_r = \omega_l - \omega_m + \omega_n \quad k_r = k_l - k_m + k_n$$

where $j=a$ and b .

$$\begin{aligned} \varrho_{bc}^{(3)}(\omega \pm \delta) = & \pm (i/h)^3 N \frac{E_{10} E_{20}^2 J_0 J_1 \mu_{ab}^2 \mu_{bc}}{r_{bc} + i(\Delta_{bc} \pm \delta + kv)} \\ & \times \left[\frac{1}{r_b \pm i\delta} \left(\frac{1}{r_{ab} + i(\Delta_{ab} \pm \delta - kv)} + \frac{1}{r_{ab} - i(\Delta_{ab} - kv)} - \frac{1}{r_{ab} + i(\Delta_{ab} - kv)} - \frac{1}{r_{ab} - i(\Delta_{ab} \mp \delta - kv)} \right) \right. \\ & + \frac{1}{r_{ac} + i(\Delta_{ac} \pm \delta)} \left(\frac{1}{r_{ab} + i(\Delta_{ab} \pm \delta - kv)} + \frac{1}{r_{ab} + i(\Delta_{ab} + kv)} \right) - \frac{1}{r_{ac} + i\Delta_{ac}} \left(\frac{1}{r_{ab} + i(\Delta_{ab} - kv)} \right. \\ & \left. \left. + \frac{1}{r_{ab} + i(\Delta_{ab} + kv)} \right) \right] \times e^{i[(\omega \pm \delta)t + kz]}, \end{aligned} \quad (5)$$

where v is the velocity component along the z axis. The reemitted field E_r is expressed by [9]

$$E_r(\omega \pm \delta) = -i \frac{KL}{2\varepsilon_0 \mu \sqrt{\Pi}} \int e^{-(v/u)^2} [\mu_{ba} \varrho_{ab}^{(3)} + \mu_{cb} \varrho_{bc}^{(3)} + \text{c.c.}] dv, \quad (6)$$

where an optically thin absorption cell is assumed. Considering $r/ku \ll 1$ and the beating current at the photodiode is proportion to

$$A = E_r(\omega + \delta) \cdot E_1^*(\omega) + E_r^*(\omega - \delta) \cdot E_1(\omega) + \text{c.c.} \quad (7)$$

The signal of modulation transfer spectroscopy in the case of cascade three-level quantum system is given by

$$I(\delta) \propto L E_{10}^2 E_{20}^2 J_0 J_1 \mu_{ab}^2 (N/u) [S_1(\delta) + S_2(\delta)] \times e^{i\delta t} + \text{c.c.}, \quad (8)$$

$$S_1(\delta) = \sum_{j=a,b} \frac{\mu_{ab}^2}{r_j + i\delta} \left(\frac{1}{r_{ab} + i(\Delta_{ab} + \delta/2)} - \frac{1}{r_{ab} + i(\Delta_{ab} + \delta)} + \frac{1}{r_{ab} - i(\Delta_{ab} - \delta)} - \frac{1}{r_{ab} - i(\Delta_{ab} - \delta/2)} \right), \quad (9)$$

$$\begin{aligned} S_2(\delta) = & \frac{\mu_{bc}^2}{r_b + i\delta} \left(\frac{1}{\bar{r} + i(\bar{\Delta} + \delta)} - \frac{1}{\bar{r} + i(\bar{\Delta} + \delta/2)} + \frac{1}{\bar{r} - i(\bar{\Delta} - \delta/2)} - \frac{1}{\bar{r} - i(\bar{\Delta} - \delta)} \right) \\ & + \frac{\mu_{bc}^2}{r_{ac} + i(\Delta_{ac} + \delta)} \left(\frac{1}{\bar{r} + i(\bar{\Delta} + \delta)} - \frac{1}{r_{ab} + i(\Delta_{ab} + \delta)} \right) + \frac{\mu_{bc}^2}{r_{ac} + i\Delta_{ac}} \left(\frac{1}{r_{ab} + i(\Delta_{ab} + \delta/2)} - \frac{1}{\bar{r} + i(\bar{\Delta} + \delta/2)} \right) \\ & + \frac{\mu_{bc}^2}{r_{ac} - i(\Delta_{ac} - \delta)} \left(\frac{1}{r_{ab} - i(\Delta_{ab} - \delta)} - \frac{1}{\bar{r} - i(\bar{\Delta} - \delta)} \right) + \frac{\mu_{bc}^2}{r_{ac} - i\Delta_{ac}} \left(\frac{1}{\bar{r} - i(\bar{\Delta} - \delta/2)} - \frac{1}{r_{ab} - i(\Delta_{ab} - \delta/2)} \right), \end{aligned} \quad (10)$$

where $\Delta_{ab} = \omega - \omega_{ab}$, $\Delta_{ac} = 2\omega - \omega_{ac}$, $\bar{\Delta} = \omega - (\omega_{ab} + \omega_{bc})/2$, and $\bar{r} = (r_{ab} + r_{bc})/2$. $S_1(\delta)$ and $S_2(\delta)$ represent saturation absorption of the intermediate level and two-photon transition of the high lying state, respectively. Comparing with fluorescence detection for two-photon transition, there is no Doppler background problem in $S_2(\delta)$ due to the near resonance of the intermediate level. In most cases of two-photon transitions, it is resonable to suppose $|\omega_{ac}/2 - \omega_{ab}| \gg \delta$. Thus the signal centers of $S_1(\delta)$ and $S_2(\delta)$ are at the resonances of ω_{ab} and ω_{ac} , respectively. Comparing with optical heterodyne detection via NDFWM of frequency modulation by AOM, modulation transfer spectroscopy with optical phase modulation has no frequency shift for the resonance center. This kind of signal pattern is very useful for precision spectroscopy measurement and laser stabilization. The line shape is calculated from (8–10), and demonstrated in Fig. 2. Maximum slope of the signal at the resonance center is given when δ is near $0.76 r_{ac}$, shown in Fig. 3.

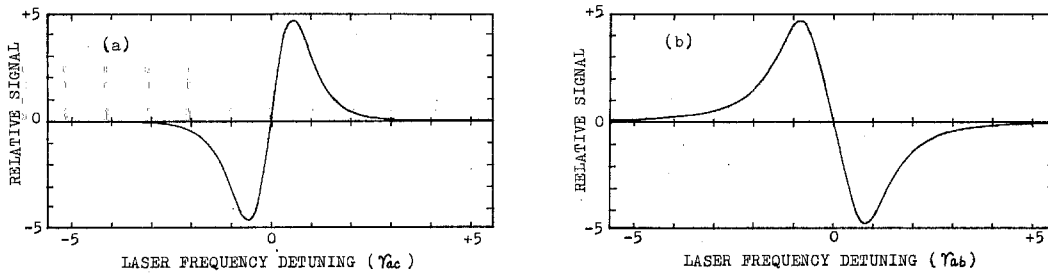


Fig. 2a, b. Line shape of modulation transfer spectroscopy via laser frequency detuning in the cases of **a** two-photon transition and **b** saturation absorption, where $\delta/r_{ac} = \delta/r_{ab} = 0.6$ and $\frac{1}{2}\omega_{ac} - \omega_{ab} = 20r_{ac}$

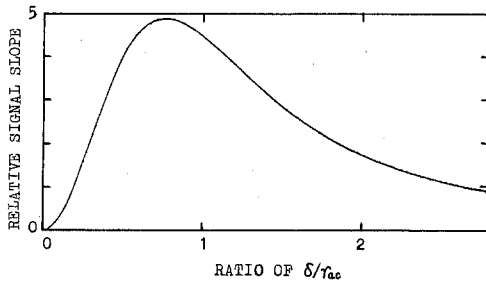


Fig. 3. Variation of signal slope at the resonance center for a two-photon transition via δ/r_{ac}

2. Experiment

The experimental configuration is shown in Fig. 4. An acousto-optic modulator is used as an optical isolator and driven by a RF power supply at 69 MHz. By rotating the $\lambda/2$ plate, the intensity ratio of pump beam to probe beam can be chosen to get maximum signal. The optical phase modulator is driven by an RF power supply at 9 MHz. As shown in figure, the modulated and unmodulated beams counterpropagate through the long arm of the cross oven containing sodium, and the polarization of the two beams are orthogonal each other. The temperature of the oven is controlled by a PID servo device, and its heating region is 25 cm long. A monochrometer combined with a photomultiplier is placed in the short arm of the oven to observe fluorescence in the direction perpendicular to the laser beams. After passing the oven, the probe beam is directed onto a photodiode by a polarizer-splitter. Optical heterodyne beating current is produced at the photodiode with frequency δ . Signals of are extracted by a double-balanced mixer. The modulated beam is chopped at 500 Hz. The leakage light from mirror M is

used as reference signal for the lock-in amplifier. Signals are recorded on a chart recorder or oscilloscope. In the tuning range of the DCM dye, we observed several two-photon transitions from the ground single state $X^1\Sigma_g^+$ to high-lying triple states of $(n)^3A_g$ [18]. The two-photon transitions are enhanced by mixing intermediate levels of $A^1\Sigma_u^+ \sim b^3\Pi_u$. A typical signal is shown in Fig. 5. Fluorescence at 365 and 680 nm are used to monitor the two-photon transition and saturation absorption, respectively. The power of pump and probe beam are 3.2 and 2.5 mW, respectively.

3. Discussion

Figure 5 demonstrates that the signal slope of a two-photon transition is opposite to that of saturation absorption, in agreement with Fig. 2. This character can be used to distinguish the signal of two-photon transition from saturation absorption. In order to compare between experimental and theoretical results, the signal of a two-photon transition shown in Fig. 5 is expanded by reducing the scan width of the laser frequency, and r_{ac} is obtained by measurement of the fluorescence signal at 365 nm. The comparison is shown in Fig. 6. It is obvious that the line shape of two-photon modulation transfer spectroscopy is in agreement with a theoretical calculation of $S_2(\delta)$. The scatter in Fig. 6 is due to laser frequency jitter and the nonlinear scan of the laser frequency. If a precise stable dye laser and a frequency scan system could be used [14], the precise measurement would be obtained. Comparing with the fluorescence signal, the modulation transfer spectroscopy has a very high sensitivity

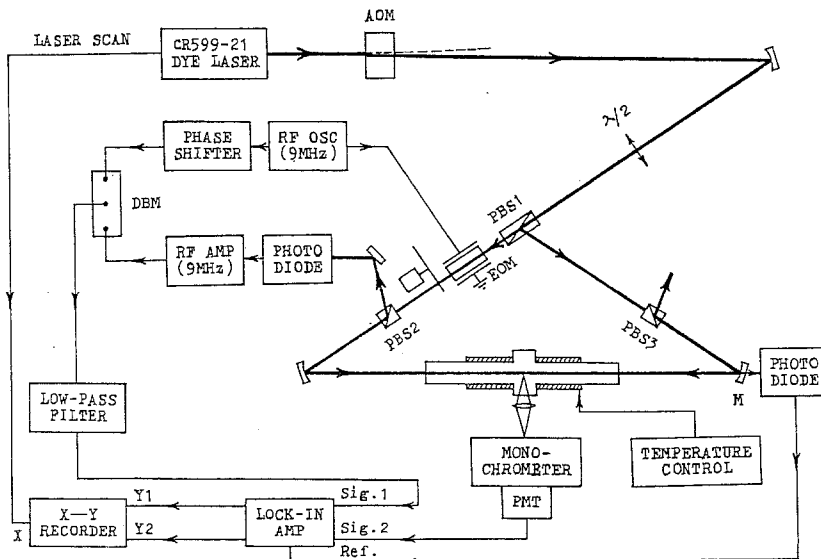


Fig. 4. Experimental configuration

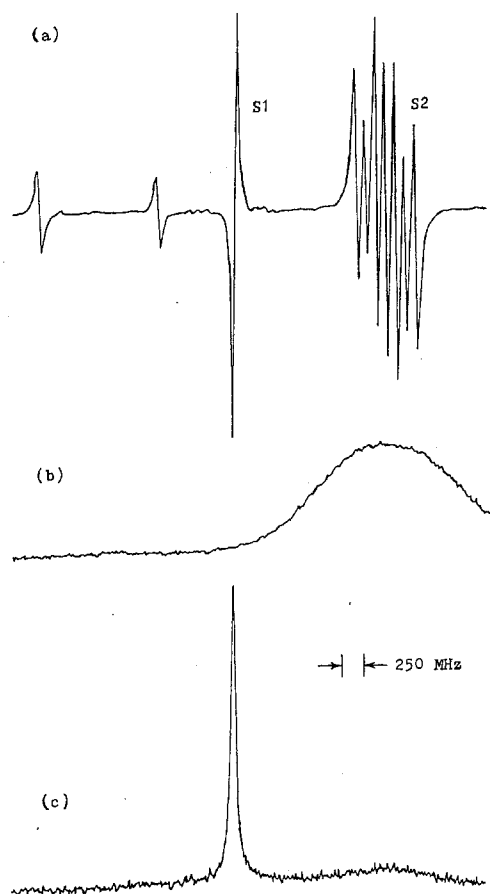


Fig. 5. **a** Optical heterodyne modulation transfer spectra of the 6436.9 Å Na_2 line taken with a modulation frequency of 9 MHz. S_1 is the two-photon transition of $X^1\Sigma_g^+ - (A^1\Sigma_u^+ \sim b^3\Pi_u) - (n)^3A_g$ and S_2 is HFS of $X^1\Sigma_g^+ - b^3\Pi_u$. **b** Fluorescence at 690 nm from $(A^1\Sigma_u^+ \sim b^3\Pi_u)$ to $X^1\Sigma_g^+$. **c** Fluorescence at 365 nm from $(n)^3A_g$ to $a^3\Sigma_u^+$. The total laser scan width is 5 GHz with the laser frequency increasing to the right

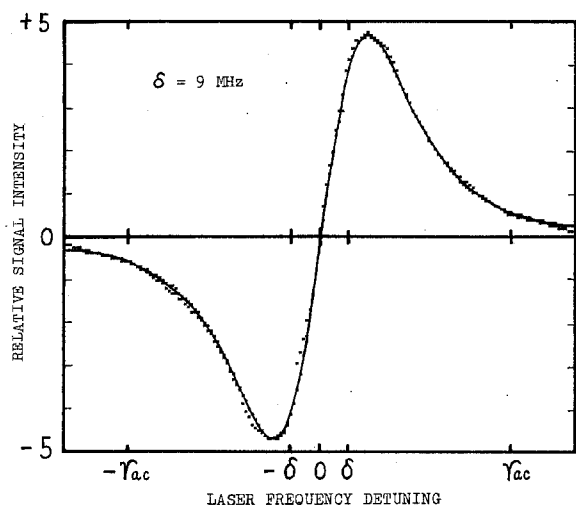


Fig. 6. Detailed signal of the two-photon modulation transfer spectroscopy in Na_2 shown in Fig. 5 and experimental profiles (·) comparing with theoretical shape (solid line) where r_{ac} is given by measurement of fluorescence signal at 365 nm

and no Doppler background problem due to the near resonance of the intermediate level. Using a fast scan and 20 μs for the RC low-pass filter, signals were observed with an oscilloscope with a S/N of 85 for the two-photon transition. The signals demonstrated in Fig. 5 were obtained with 0.1 s for the RC low-pass filter and a S/N of the fluorescence signal at 365 nm was found to be 72. So it is easy to predict that the S/N of the fluorescence signal approximates to 1 with 20 μs for the RC low-pass filter. Figures 6 and 3 demonstrate that if a suitable modulation frequency δ is found, the signal of the two-photon modulation transfer spectroscopy in Na_2 can be further increased and therefore the S/N in two-photon transition can be 200 times higher than the fluorescence detection in our experimental set. Its sensitivity could be further enhanced by an external optic ring cavity containing the sample cell [19]. We expect to study high-lying states of molecules with two-photon modulation transfer spectroscopy.

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